

DriftGuardian Technical Report: Architecture and Use Case Alignment

1. Use Case: A Governance Gap Before SOP Publication

Current enterprise AI governance tools often focus on model monitoring, prompt safety, or general content review, but they rarely check whether an AI-generated operational SOP still conforms to approved internal policies before publication. This creates a serious gap in regulated workflows: a small change in a KYC SOP, such as weakening a risk threshold, replacing an approval role, omitting a mandatory step, or extending an escalation window, can directly become a compliance failure.

DriftGuardian addresses this gap as a pre-deployment governance gate for AI-generated SOPs. The current prototype focuses on KYC/AML because the procedures are rule-based, high-risk, and business-critical. Instead of reviewing writing quality, the system checks whether the SOP is safe to release into operations.

2. Innovation: From Simple Comparison to Policy-Hierarchy Conformance

A major limitation of existing document comparison or RAG-based review tools is that they treat all differences as suspicious. This does not match real enterprise policy structures, where regional policies may legitimately override global baselines.

DriftGuardian introduces policy-hierarchy conformance checking: Global Baseline → Approved Regional Override → AI-Generated SOP. Regional rules override global rules only where explicitly approved; otherwise, the global baseline remains binding. This allows the system to detect unauthorized drift while avoiding false positives from legitimate regional variation. The output is a clear operational decision: PASS, WARN, or BLOCK.

3. Architecture: OPEA-Style Modular Governance Pipeline

DriftGuardian contains the following main components:

Component	Implementation	Role
Reviewer UI	Flask frontend	Upload global policy, regional override, and SOP; display verdict, findings, coverage, payloads, and audit record
Megaservice gateway	FastAPI /v1/drift_check	Exposes the main validation endpoint and coordinates the backend workflow

Document loader	drift/loader.py	Extracts text from PDF, DOCX, TXT, and Markdown files
Drift engine	drift/engine.py	Runs a two-stage LLM pipeline for policy requirement extraction and SOP verification
LLM endpoint	OpenAI-compatible local service	Uses a local model endpoint, with Ollama as the default demo backend and TGI-compatible deployment as a production path
Evaluation harness	evaluation.py	Runs labelled demo cases and reports verdict accuracy, latency, and unsupported-claim rate
Deployment	Docker Compose / deploy script	Starts backend, UI, and LLM service on a single-node environment

The architecture follows an OPEA GenAIEXamples-style pattern: a thin API gateway coordinates modular services, the LLM is exposed as a replaceable inference microservice, and the stack is containerized for reproducible deployment. This makes the prototype suitable for single-node evaluation while preserving a migration path toward a fuller OPEA deployment with standard GenAIComps and hardware-optimised inference services.

4. Enterprise Output: From AI Feedback to Remediation Workflow

Another market shortcoming is that many AI review tools produce unstructured comments that still require manual interpretation. DriftGuardian produces enterprise-ready outputs: detailed findings, remediation actions, audit metadata, document hashes, and Jira/Confluence-ready payloads. This means a blocked SOP can immediately become a trackable compliance task, while audit records preserve evidence of what was checked, when, and against which source documents.

5. Market Potential: Scalable Control for AI-Generated Operations

As AI-assisted drafting becomes common, regulated organizations need scalable guardrails for controlled documents. DriftGuardian's value is practical: it reduces manual review burden, prioritizes human compliance attention, and blocks high-risk policy drift before production.

Although the prototype focuses on KYC/AML, the same architecture can extend to healthcare SOPs, manufacturing quality procedures, privacy policies, procurement rules, and internal security controls.